Optimizing Cuckoo Search using Genetic Operators and Correlation-Aware Fitness for Plant Leaf Classification

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CCS → Machine learning → Machine learning approaches → Bio-inspired approaches → Evolutionary algorithms

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Algorithms, Performance, Experimentation, Measurement, Design

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Binary Cuckoo Search Algorithm; Plant Classification; Feature Selection; Ensemble Learning; Factorization Machine

1. INTRODUCTION

1.1 Background of the Study

The large volume of information generated during feature extraction remains a significant challenge for leaf classification in precision agriculture. With thousands of texture, shape, and color features, many of which are irrelevant or redundant [1], dimensionality reduction is an established problem. The literature on leaf classification up to 2025 has presented positive results with feature fusion [2], but at the expense of increased complexity. Therefore, the need to find more effective methods to reduce dimensionality is clear.

The resemblance between leaves is another significant factor in leaf classification, making the development of effective, easy-to-interpret models a priority. The classification models tend to overfit and generalize poorly, requiring dimensionality reduction using specialized techniques. With the metaheuristic algorithm Binary Cuckoo Search (BCS), researchers demonstrated that the wrapper-based method is effective in finding the optimal feature set.

Researchers found that feature optimization is key to classifying regional flora, particularly Philippine Medicinal Plants. The Philippines is recognized by scientists as a biodiversity hotspot, with many herbal plants such as Banaba (Lagerstroemia speciosa) and Lagundi (Vitex negundo).

The plants have exhibited complex morphological similarities that pose challenges comparable to those in global benchmarks. However, the usual classifiers have not been effective at distinguishing among various categories of medicinal plants in the region due to considerable feature overlap. In this regard, the present study proposes that the model does not follow a particular architectural principle. The model incorporates Inception V3 for feature extraction. In addition, GBCS optimization helps reduce noise and redundancy in the features of Philippine medicinal plants. Researchers suggest that the model's capabilities can extend its applicability for the precise identification of medicinal plants in the region, compared to current classifiers like Flavia and Swedish.

The Genetic Operators feature is an innovation that addresses the limitations of traditional swarm intelligence. The use of crossover and mutation operators to prevent premature convergence during the optimization process improves the model.

In this regard, the Cuckoo Search algorithm can be allowed to perform its usual Lévy flight optimization. Subsequently, researchers can use genetic operators to incorporate the characteristics of successful solutions in the population. According to the theoretical analysis in [3], the probability of the operators plays a critical role in balancing exploration. In this regard, researchers can consider the model effective in preventing premature convergence. The model demonstrates adequate exploration, aligning with the criteria in [4], [5].

A Correlation-Aware Fitness Function is a specific evaluation strategy developed to optimize the compactness and independence of the selected features. Researchers should note that while traditional fitness functions seek to enhance classification accuracy and the number of features, they often fail to account for feature relationships. Methods that ignore feature relationships commonly incorporate redundant, highly correlated features that provide no unique value to the classifier [6], [7]. The methodology proposed in this paper introduces a correlation penalty into the objective function. The penalty ensures that the selected features have their redundancy minimized and that each provides unique information to the classifier. The classifier will therefore be able to generalize more efficiently with a reduced and more efficient dataset.

The existing literature shows that incorporating Genetic Operators into the optimization process yields better results than the conventional Cuckoo Search algorithm in complex scenarios [8], validating the effectiveness of the proposed integration.

Premature convergence, a substantial drawback of the conventional BCS algorithm, has been effectively mitigated by mutation operators in validation studies [9]. Population diversity was ensured during the modeling process by the methodological framework, promoting convergence to the global optimum rather than local optima.

Additionally, the fundamental research validates the effectiveness of the BCS model, thus validating the selection of the baseline model for plant leaf classification. The baseline model shows that while the standard model performs well, it also reveals its limitations, which this study aims to improve upon. The findings show that the conventional model can stagnate, and the fitness function largely ignores feature dependencies. Correlation-Aware Fitness Function demonstrates the importance of the hybrid model for improved performance on biological datasets.

Lastly, to demonstrate the effectiveness of the correlation measures used in the fitness function, recent research [10] has shown that using correlation analysis significantly improves the efficiency of the feature selection process. The findings show that while the accuracy-based model selects features for prediction, incorporating awareness of correlations and eliminating redundant features significantly improves the feature set. The findings demonstrated that eliminating features improves feature simplicity and generalizability of the classifier, producing a smaller feature set for better computational efficiency.

Table 1 below presents the key studies used in this research, contrasting the baseline model proposed by Al-Batah et al. with recent improvements to the algorithm for better performance.

Table 1. Summary of Related Literature and Identification of Research Gaps

Ref. No	Author(s) & Year	Domain/Focus	Methodology Used	Key Findings	Identified Limitations & Relevance to Current Study
[1]	Al-Batah et al. (2025)	Plant Leaf Classification (Baseline Study)	Binary Cuckoo Search (BCS) with Factorization Machines.	Achieved 99.6% accuracy on the Flavia dataset. Demonstrated that wrapper-based FS outperforms filter-only methods.	The authors explicitly note that the algorithm suffers from premature convergence, and the fitness function ignores feature correlation.
[2]	Li et al. (2025)	Virtual Power Plant Optimization	Genetic-Heuristic Fusion Algorithm (GA + Ant Colony).	Demonstrated that combining global search (GA) with local optimization prevents the "premature convergence" typical in single-algorithm approaches.	Validates that hybridizing Genetic Operators is a current, state-of-the-art strategy (2025) to fix convergence issues, even in complex resource allocation tasks.
[3]	Sun & Su (2022)	Adaptive Genetic Algorithms	Adaptive GA with Nonlinear Probabilities.	Mathematically analyzed how adjusting crossover/mutation rates balances exploration vs. exploitation.	Provides the mathematical justification for using dynamic genetic operators to solve the "premature convergence" problem.
[4]	El-Shorbagy et al. (2024)	Engineering Optimization	Hybrid Monarch Butterfly Optimization + GA Operators.	Proved that embedding GA crossover into swarm algorithms improves solution stability.	Confirms that hybridizing swarm intelligence with genetic operators is a valid, modern strategy to fix local optima issues.

[5]	Li et al. (2020)	Logistics Optimization	Cuckoo Search + Genetic Operators.	Prove that genetic operators allow the Cuckoo algorithm to "jump" out of local optima.	Empirical proof that this specific hybrid (CS + GA) works for preventing stagnation.
[6]	Hall (1999)	Machine Learning Theory	Correlation-based Feature Selection (CFS).	Established the heuristic that "good feature subsets contain features highly correlated with the class, yet uncorrelated with each other."	Serves as the theoretical basis for the proposed "Correlation-Aware Fitness Function."
[7]	Xie et al. (2022)	Microarray Data Classification	Binary Differential Evolution + Correlation Fitness.	Demonstrated that penalizing correlation in the fitness function reduced subset size without hurting accuracy.	Proves that modifying the fitness function to penalize correlation creates more efficient models.
[8]	Angelova et al. (2023)	Parameter Identification	Multi-population GA + Cuckoo Search.	The hybrid approach showed superior stability compared to standard CS.	Further validates that mixing evolutionary strategies (GA) with swarm intelligence (CS) improves exploration.
[9]	Xie et al. (2022)	Image Matching / Optimization	Directed Crossover Genetic Algorithm (DCGA).	Proposed a "multilayer mutation" strategy that improved matching accuracy by maintaining diversity.	Supports the use of advanced mutation strategies specifically for high-dimensional visual data.
[10]	Chen et al. (2025)	Fault Diagnosis	Correlation-Guided Particle Swarm Optimization (PSO).	Used correlation to guide the search, resulting in smaller, distinct feature subsets.	Highlights the necessity of Correlation-Awareness in swarm algorithms to handle high-dimensional data.

The literature review has clearly outlined the trajectory of research on feature selection, including the potential and limitations of metaheuristic algorithms in this area. Although the seminal research in this area has clearly outlined a robust framework for classifying plant leaves using Binary Cuckoo Search, it acknowledged that premature convergence and redundancy in the selected features, due to inadequate analysis of correlations, were significant issues that researchers had yet to solve.

To address the identified shortcomings in this area of research, there are evidently feasible research methodologies that researchers can utilize. For instance, the mathematical justifications outlined in research studies clearly indicate that the inclusion of Genetic Operators in this area of research is vital to avoid stagnation in the optimization process. On the other hand, research studies in this area clearly indicate that the inclusion of correlation effectively eliminates redundancy in the selected features, and this study aligns with the recommendations outlined in the baseline research on Plant Leaf Classification.

1.2 Purpose and Description

The current study aims to improve the Basic Binary Cuckoo Search (BCS) optimization algorithm by developing a new variant, Genetic Binary Cuckoo Search (GBCS). This optimization technique aims to solve the current challenges in local search convergence and variable interdependencies in optimization algorithms. Moreover, the GBCS optimization technique plans to improve global search and minimize data redundancy in a "waterfall" system by using crossover and mutation operations in genetic algorithms. The proposed optimization technique includes acquiring leaf images, extracting high-dimensional features with Inception V3, and using GBCS to determine an optimal low-correlation feature set that accurately classifies plant species. Scientists in biodiversity studies and agricultural fields can further utilize the derived high-precision optimization method. The contribution to achieving the United Nations Sustainable Development Goals (SDGs) on Life on Land (SDG-15), Zero Hunger (SDG-2), Good Health and Well-being (SDG-3), and Climate Action

(SDG-13) is directly attributable to the utilization of the suggested optimization method.

Researchers carry out this process through a modular pipeline. This process begins with Image Acquisition and Feature Extraction, where researchers use deep learning techniques to extract features. GBCS Feature Selection, a significant feature of the process, improves feature subsets by incorporating solution properties to reduce redundancy.

The process ensures that the last process used unique features. Comparative analyses of accuracy, dimensionality reduction, and convergence rates of GBCS, BCS, and other leading techniques using Swedish, Flavia, and Philippine Medicinal Leaf Datasets validate the process.

1.3 Objectives

1.3.1 *General Objectives*

To optimize the efficiency of the Genetic Binary Cuckoo Search (GBCS) algorithm based on the principles of evolutionary diversity and the use of a correlation-penalized evaluation strategy. The GBCS aims to prevent premature convergence and feature redundancy for high-dimensional plant leaf classification.

1.3.2 Specific Objectives

1.3.2.1 To extract high-dimensional Inception-V3 feature embeddings $|F| = 2{,}048$ for the Swedish, Flavia, and Philippine plant leaf datasets as direct input for wrapper-based optimization..

1.3.2.2 To optimize the performance of the Binary Cuckoo Search algorithm by proposing the Genetic Binary Cuckoo Search (GBCS) framework, which incorporates Uniform Crossover and Bit Flip Mutation operators and integrates them with the Lévy flight strategy. The Correlation-Aware Fitness Function will incorporate the Pearson Correlation Coefficient.

1.3.2.3 To compare the performance of the GBCS model against the baseline Binary Cuckoo Search and other state-of-the-art metaheuristics in addressing the Swedish and Flavia benchmark data sets, in terms of feature reduction ratio, rate of convergence, classification accuracy, and stability of the algorithm.

1.3.2.4 To compare the performance of the GBCS model in addressing the Swedish, Flavia, and Philippine Medicinal Plant Leaf data sets, thus evaluating the GBCS model's ability to generalize and perform across local botanical variations by analyzing the differences in data sets.

1.4 Scope and Limitations

This research aims to optimize the Genetic Binary Cuckoo Search (GBCS) algorithm for solving the NP-hard Feature Selection (FS) problem, particularly for plant leaf feature selection datasets with an ample feature space. The research is limited to addressing and optimizing the standard Binary Cuckoo Search framework using a dual-optimization approach. The algorithm includes incorporating Genetic Operators, Uniform Crossover, and Bit Flip Mutation to enhance global search. Furthermore, it includes developing a Correlation-Aware Fitness Function based on Pearson's Correlation Coefficient for addressing feature redundancy. Additionally, it is limited to solving and addressing the problem of plant species identification. For evaluation, researchers used two datasets: the Flavia dataset, consisting of 1,907 images of 32 plant species, and the Swedish Leaf dataset, comprising 1,125 images of 15 plant species. Furthermore, it is limited to using Inception V3, a pre-trained Convolutional Neural Network (CNN), solely for extracting features, not for fine-tuning its internal CNN layers to generate high-dimensional feature vectors. Lastly, the performance of the proposed model is evaluated and compared with the standard Binary Cuckoo Search framework proposed by Al-Batah et al. (2025).

The feature selection process also has certain limitations. The GBCS algorithm, a wrapper-based feature selection technique, uses the k-NN classifier internally to evaluate feature subsets. The GBCS then used the evaluated feature subset for the classification with the Factorization Machine. The overall process makes the GBCS algorithm computationally expensive. Therefore, the study focuses more on the classification accuracy and the compactness of the feature subsets rather than the execution speed of the GBCS algorithm.

Additionally, the proposed GBCS algorithm focuses specifically on the classification of individual leaf images acquired under simple or uniform backgrounds, as used in the benchmark datasets. The proposed GBCS algorithm does not address the classification and detection of plants in diverse, complex outdoor environments with overlapping leaf structures and varying lighting conditions. Lastly, the proposed Correlation-Aware Fitness Function uses Pearson's Correlation Coefficient to identify linear correlations between features. The proposed GBCS algorithm, consequently, focuses on eliminating linear correlations and dependencies among features, and thus cannot account for nonlinear correlations and dependencies in the high-dimensional feature space.

2. METHODOLOGY

The present study employed the Cross Industry Standard Process for Data Mining (CRISP-DM) methodology to develop and validate the Genetic Binary Cuckoo Search (GBCS) model. The CRISP-DM methodology is a data mining process model that ensures the development of a feature selection model that is both theoretically sound and practically applicable. The methods employed in the present study comprise six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.

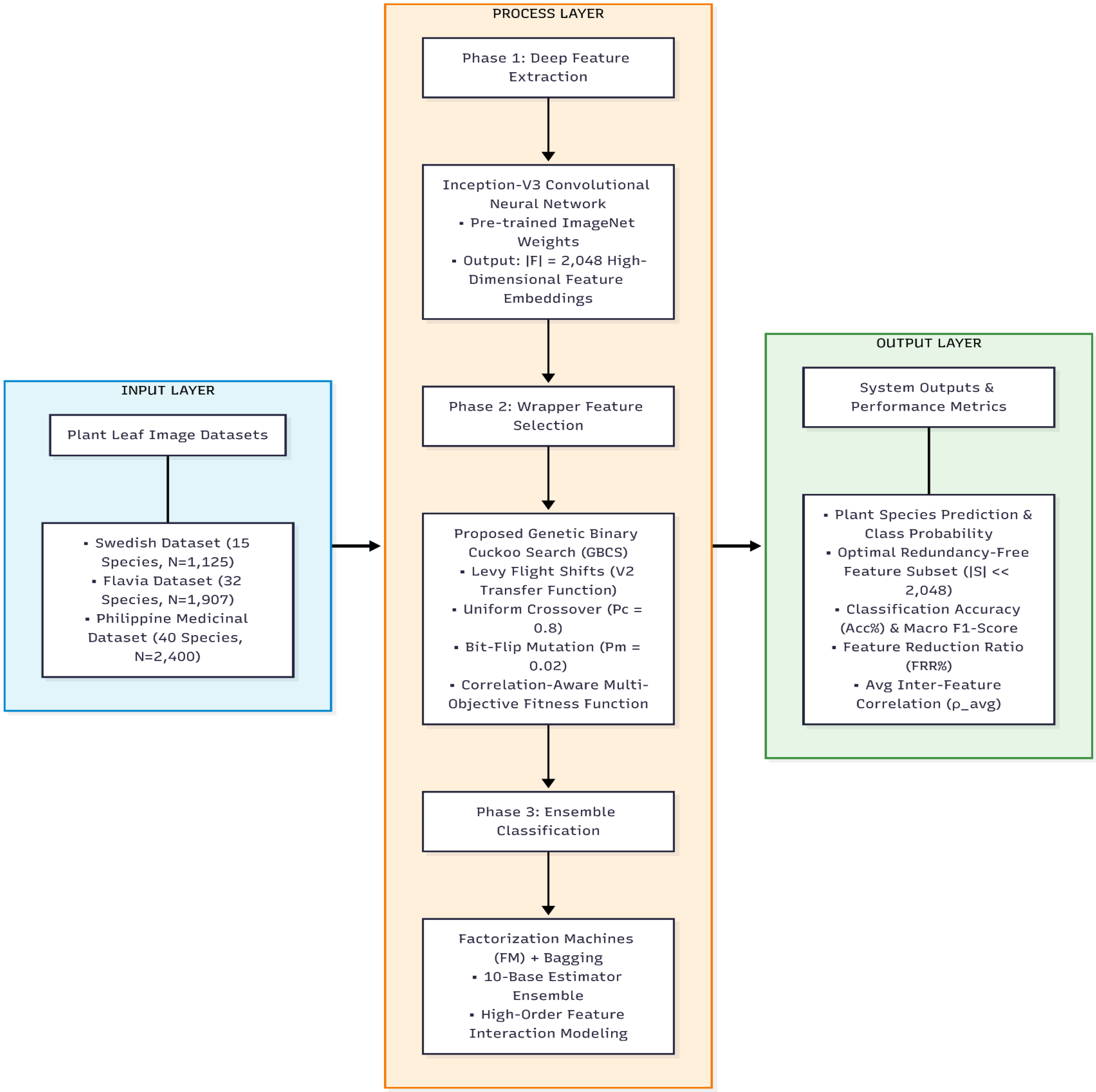

Figure 1. Conceptual Framework

2.1 Phase 1: Business Understanding

The first phase of the methodology employed in the present study involves articulating the research objectives and identifying the data requirements for developing the automated plant identification system. The primary purpose of the proposed model is to create a classification system that addresses the curse of dimensionality in ethnomedicinal leaf datasets, where overlapping morphological features often compromise the classification of plant species.

2.1.1 Benchmark Datasets

To evaluate the generalizability of the developed model, researchers chose two existing benchmark datasets as control groups for this study. The Swedish Leaf Dataset, with its balanced set of 1,125 images of 15 different tree species, is selected as the primary control group. The Flavia Leaf Dataset, comprising 1,907 images of 32 species, is selected as the secondary control group for this study.

2.1.2 Experimental Dataset

The primary dataset used in this paper is the Philippine Medicinal Plant Leaf Dataset (PMLP) with local ethnomedicinal identifications. The dataset consists of 40 unique image classes totalling 4,971 images. The difference between this dataset and the control datasets is that it is highly imbalanced and also presents a high level of morphological similarity between classes. The selection makes it a suitable dataset for testing the Genetic Binary Cuckoo Search (GBCS) in its capacity to handle unstructured biological information, which is known to be difficult for traditional algorithms.

2.2 Phase 2: Data Understanding

After acquiring the datasets, the second phase of GBCS is Data Understanding, which assesses the quality and statistical properties of the datasets.

2.2.1 Exploratory Data Analysis (EDA)

Researchers conducted an exploratory data analysis of the collected datasets to assess the structural integrity and the complexity of class distribution in the images. Exploratory data analysis is crucial for defining the complexity of the classification problem and emphasizing the need for feature selection in GBCS.

2.2.2. Dataset Structure and Preprocessing

The researchers highlighted the variability in the dataset structure in the structural analysis. In other words, the Swedish Leaf Dataset, which had 1,125 images, and the Flavia Leaf Dataset, which had 1,907 images, had a flat directory structure. However, the researchers addressed this by a metadata-driven approach that enabled the transformation of the datasets into a structured, labeled CSV format. On the other hand, the Philippine Medicinal Plant Leaf Dataset (PMLD), which had 4,791 images, had a hierarchical directory structure. Initially, preprocessing was carried out to remove the effects of the directory naming structure, and this led to a clean dataset with 40 unique ethnomedicinal plant species and valid image references. It was also noticed that feature extraction was carried out successfully, which validated the use of the Inception V3 architecture in that it produced 2,048-dimensional feature vectors (n_0, n_1, …, n_{2047}).

2.2.3. Comparative Class Distribution

As depicted from the class distribution, the classification difficulty varies, as depicted in Figure 2. The variation is evident when the control and experimental group datasets used in the experiment, and their corresponding class distribution are considered.

2.2.3.1 Swedish Dataset (Control Group 1): This dataset is perfectly balanced, consisting of exactly 15 classes with 75 images per class. This dataset serves as an optimal baseline for validating the stability of the algorithms.

2.2.3.2 Flavia Dataset (Control Group 2): The researchers found this dataset to have a slight class imbalance after consolidating the effects of duplicate class labels, such as "Crape myrtle."

2.2.3.3 Philippine Dataset (Experimental Group): The dataset clearly shows a class imbalance issue, which is expected in natural settings and is a hallmark of biological datasets. The classes are imbalanced, ranging from a minimum of 100 images of "Curcuma longa" to a maximum of 171 images of "Tamarindus indica."

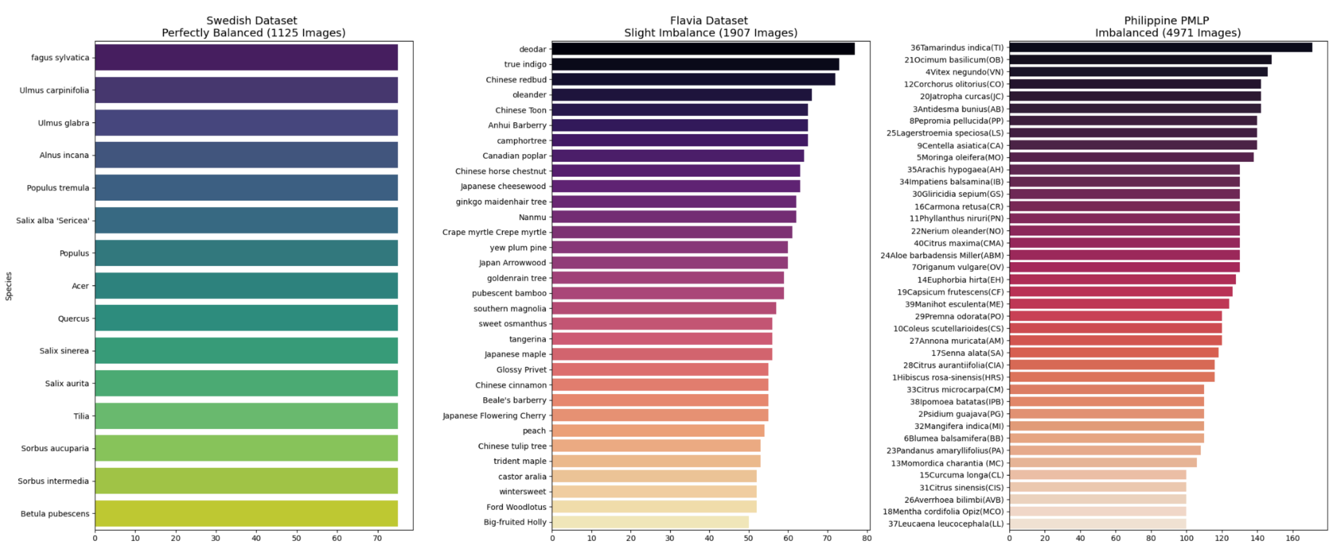

Figure 2. EDA of the Three Datasets

2.2.3.4 Imbalance and Morphological Analysis

Analysis of the Philippine dataset shows it is imbalanced by a factor of 1:1.71. The Philippine dataset is not extremely imbalanced and does not need synthetic oversampling techniques such as SMOTE. However, it is evident that the larger classes, such as "Tamarindus indica," have almost twice as many training examples as the smaller classes. The EDA validates the approach and emphasizes the need for a feature selection technique that works well without biasing towards larger classes.

The morphological analysis shown in Figure 2 clearly illustrates "The Curse of Dimensionality." The results clearly show that similar classes are highly comparable. For instance, Citrus species such as "C. microcarpa," "C. aurantiifolia," and "C. sinensis" are highly similar in terms of venation and edge serrations. EDA clearly illustrates the need to implement the Correlation-Aware Fitness Function, eliminate redundant features (such as texture), and focus on unique features (such as morphology).

2.3 Phase 3: Data Preparation

The purpose of the data preparation phase is to transform the raw images of the leaves into a structured numerical form. This section identifies the set of tools used to incorporate the unstructured image data with the algorithmic input data.

2.3.1 Inception V3

To transform the raw visual data into a structured numerical form, the research makes use of the Inception V3 convolutional neural network. Inception V3 is a pre-trained model that can be used for the detection of complex patterns, textures, and geometric arrangements in images. In the research, the "Image Embedding" widget in the Orange Data Mining software environment feeds images of leaves into the Inception V3 model. Unlike the usual image classification task that makes use of the output of the last pooling layer in the Inception model, the research makes use of the output of the last pooling layer to transform the images into a high-dimensional numerical vector.

2.3.2 Data Partitioning

The researchers ensure statistical validity and comparability with the existing literature, and the experimental design incorporates specific data partitioning schemes. For the control experiments, the data partitioning protocols conform to standard norms. The Swedish Leaf Dataset has 25 images per class for training and 50 for testing, while the Flavia Leaf Dataset has 40 images per class for training and 10 for testing. For the Philippine Medicinal Plant Leaf Dataset (PMLP), researchers used Stratified Random Sampling with an 80:20 ratio as the data partitioning protocol for class balance. This data partitioning protocol uses 80% of the data for training ($n_{3,976}$) to optimize the performance of the model, while 20% is used for testing (n_{995}). This protocol maintains the proportion of the 40 ethnomedicinal plant species, including the minority class Curcuma longa, in the training and test sets to eliminate sampling biases.

2.4 Phase 4: Modeling

This phase deals with modeling and its development. The baseline architecture compares the model to itself to validate the proposed approach.

2.4.1 Baseline Architecture (BCS)

The Baseline Binary Cuckoo Search is used as a baseline algorithm for this experiment. This approach is a basic Binary Cuckoo Search that incorporates Lévy flights for global search and a simple random walk for local exploitation. The fitness function is optimized for accuracy and feature set size, but does not account for feature redundancy. The approach is also a sequential model in which the search function identifies a feature set based on the number of features and classification accuracy.

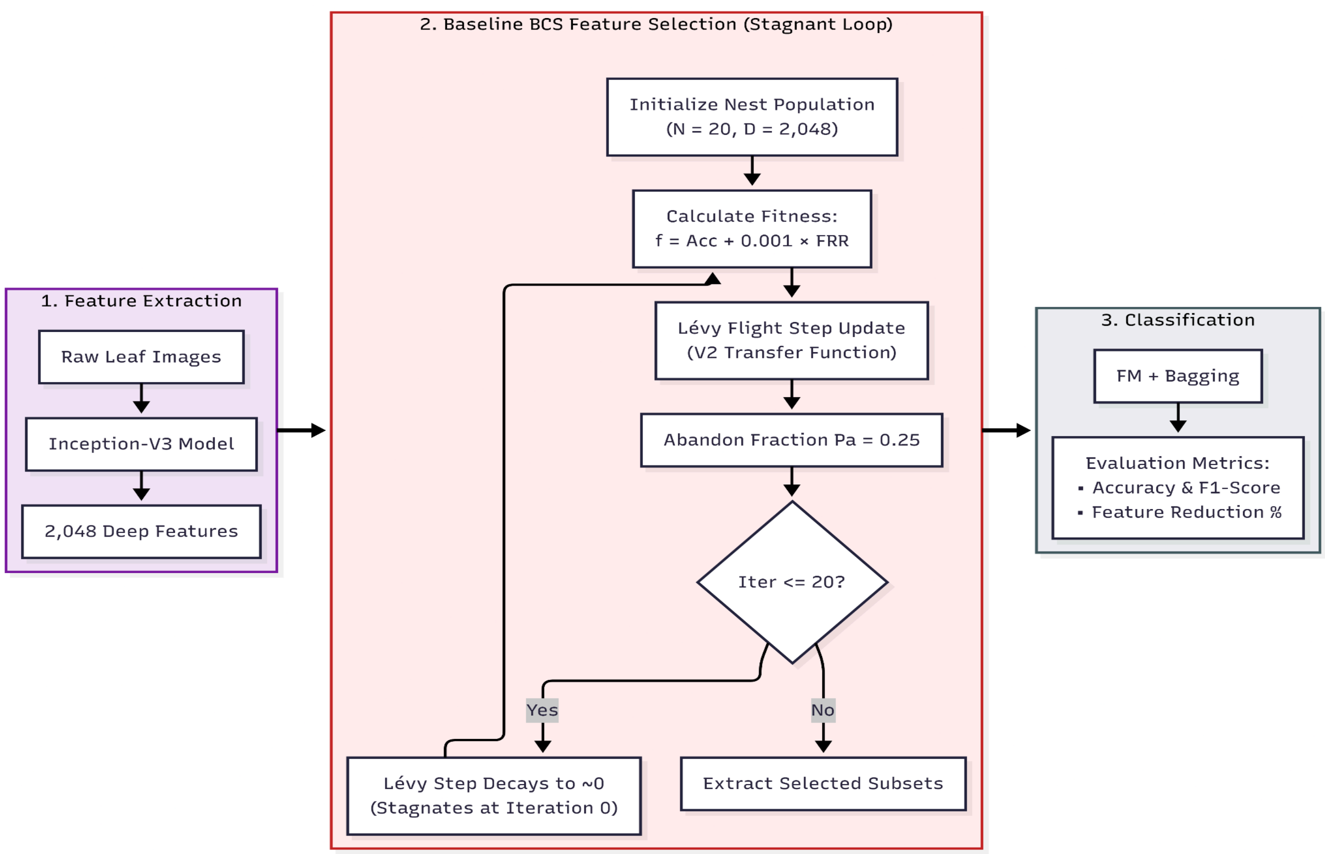

Figure 3. Baseline's Binary Cuckoo Search Architecture

2.4.2 Proposed Architecture (GBCS)

The Proposed Genetic Binary Cuckoo Search approach presents a unified hybrid algorithmic framework addressing the disadvantages of the baseline approach. Unlike traditional ways, in which algorithms are sequentially applied in a series of steps, GBCS presents a unified approach in which Genetic Operators are incorporated into the Cuckoo Search iteration loop.

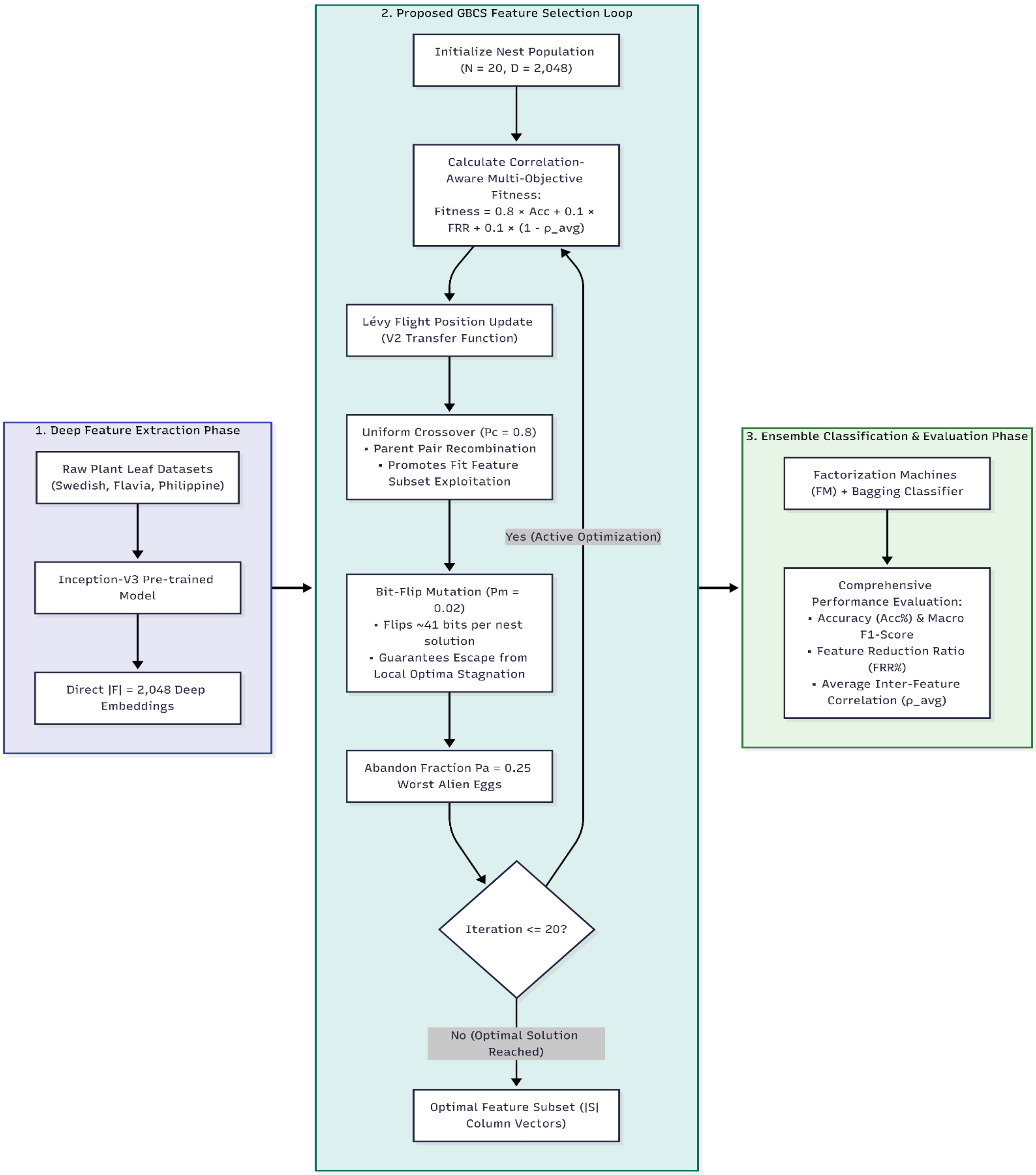

Figure 4. Proposed Genetic Binary Cuckoo Search Architecture

2.4.3 Hybrid Search Mechanism

The approach begins with a set of nest populations, represented as binary feature vectors. Lévy flights are incorporated as a global search mechanism that enables significant movement within the solution space. Following this, genetic operations are applied to the populations of nests. The approach incorporates a Uniform Crossover Operator between pairs of high-quality nests. The application of a Bit Flip Mutation Operator is as follows. This mutation is critical because it generates much-needed variation within nest populations, preventing stagnation and premature convergence in the Cuckoo Search algorithm.

2.4.4 Correlation-Aware Fitness Function

The primary mathematical optimization technique utilized in this study is the Correlation-Aware Fitness Function. The traditional wrapper-based technique considers feature subsets based on prediction accuracy and subset size. However, this technique often results in selecting redundant features, which are features that contain overlapping information. For instance, two different feature vectors could potentially carry similar information regarding texture patterns. To avoid such redundancy in the selected set of features, GBCS incorporates a statistical penalty for correlations between features. The fitness of each candidate solution, or Nest, is calculated by the resulting weighted multi-objective function:

$$Fitness = w_1 \cdot Acc + w_2 \cdot \left(1 - \frac{|S|}{|F|}\right) + w_3 \cdot (1 - \rho_{avg})$$

In this context, Acc represents the classification accuracy obtained via the k-NN wrapper approach on the selected set of features using the feature mask approach. In this equation, |S| represents the number of features in the selected set of features using the feature mask approach. In this equation, |F| represents the total number of features available in the extracted dataset (|F| = 2,048 raw features from Inception-V3). In this context, ρavg is the average Pearson correlation coefficient calculated on all pairs of features in the selected set of features.

The parameters w_1, w_2, and w_3 are set equal to 0.8, 0.1, and 0.1, respectively. The configuration must be comparable to the baseline approach and provide a fair comparison between the baseline and GBCS algorithms. The parameter values are set by the researchers so that the weight on subset size remains relatively small compared to those on accuracy and correlation, as in the baseline approach (w_2 = 0.1). The weight on accuracy decreased slightly (w_1 = 0.8) to account for the additional objective of minimizing feature correlations (w_3 = 0.1). The term $(1 - \rho_{avg})$ is known as the independence reward function. When features are highly correlated with each other (ρ = 1), this term is set equal to 0, and when features are independent of each other (ρ ≈ 0), this term is set equal to 1.

2.4.4.1 Sample Calculation (Philippine Banaba Leaf)

To illustrate the process of inputting the dataset into the equation, a sample calculation is demonstrated using a dataset from the Philippine medicinal plant leaf dataset. Let us consider an example of an image of Lagerstroemia speciosa, commonly known as Banaba. After passing through the Inception V3 feature extractor and the mRMR filter, the features are reduced to five attributes. The normalized feature values are as follows: f_1 = 0.85 for Venation contrast, f_2 = 0.84 for Venation density (which is visually redundant with f_1), f_3 = 0.12 for Aspect ratio, f_4 = 0.90 for Green intensity, and f_5 = 0.15 for Edge serration. After passing through the GBCS, the binary solution vector, Nest, is calculated as X_i = [1, 1, 0, 1, 0]. This indicates that features f_1, f_2, and f_4 are selected, while features f_3 and f_5 are not. The fitness assessment happens at four stages. First, the wrapper accuracy for the classification will be found using the k-NN classifier. Since the features used are {f_1, f_2, f_4}, the classification accuracy is 0.94. Thus, Acc = 0.94. Next, the model calculates the feature reduction ratio. From 5 features, the feature reduction ratio reduced it to 3 features; thus, the score is 1 - (3/5) = 0.40. Next, the model calculates the correlation penalty. In this case, there are only three features, and the model calculates the pairwise correlation. For instance, a correlation coefficient of 0.95 represents a high level of redundancy between f_1 and f_2. However, a correlation coefficient of 0.10 indicates a low level of redundancy between f_1 and f_4, and a similar pattern holds for f_2 and f_4 (r = 0.15). The average correlation is (ρ_{avg}) = (0.95 + 0.10 + 0.15) / 3. Thus, the independence score is 1– 0.40 = 0.60. Lastly, the total fitness is calculated using the weights defined above (w_1 = 0.8, w_2 = 0.1)

$$Fitness = (0.8 \times 0.94) + (0.1 \times 0.40) + (0.1 \times 0.60)$$

$$Fitness = 0.752 + 0.040 + 0.060$$

$$Fitness = 0.852$$

Interpretation:

The high correlation between f_1 and f_2 (0.95) significantly affects the correlation component. If the mutation removes the feature f_2 and leaves only f_1 and f_4, the correlation component will be zero, and the feature reduction score will increase, thereby improving the fitness function. The mathematical model ensures the evolution of GBCS towards the optimum features without redundancy.

2.4.5 Factorization Machine (FM)

The study uses the Factorization Machine (FM), a classification-based machine learning model that combines the benefits of Support Vector Machines (SVMs) and factorization. The model is used for its ability to handle interactions among variables, including those in sparse data, thereby enabling it to handle high-dimensional features. The model is used to obtain accurate predictions of plant species while maintaining linear computational complexity by using factorized parameters to capture variable interactions.

2.5 Phase 5: Evaluation

The evaluation phase assesses the effectiveness and efficiency of the proposed Genetic Binary Cuckoo Search (GBCS) algorithm relative to the existing Binary Cuckoo Search (BCS) algorithm. During this phase, extensive quantitative analysis is performed to evaluate the effectiveness of the proposed model in achieving the desired balance between exploration and exploitation using the integration of genetic operators. Each analysis will be discussed separately to ensure a comprehensive evaluation of the proposed model with respect to the Philippine Medicinal Plant Leaf Dataset.

2.5.1 Classification Accuracy

Classification accuracy is the ratio of correctly predicted values to the total number of values. It represents the most direct measure of the model's prediction capability. While the primary goal of this research is optimization, classification accuracy functions as a vital validation constraint, which ensures that the model has not compromised its ability to classify different shapes of leaves due to the reduction of features. The classification accuracy formula is:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

In evaluating the Lagerstroemia speciosa (Banaba) class in the Philippine data set, the model accurately identified 23 true positives (TP) and 965 true negatives (TN). However, it also produced 5 false positives (FP) and 2 false negatives (FN). Plugging in the correct figures in the formula:

$$Accuracy = \frac{23 + 965}{23 + 965 + 5 + 2}$$

$$Accuracy = \frac{988}{995}$$

$$Accuracy = 99.29\%$$

2.5.2 Convergence Speed

The speed of convergence is another significant parameter that measures the efficiency of the optimization process. It refers to the number of iterations required before the optimization algorithm reaches its optimal fitness value or stabilizes. In this research, the rate of convergence was used as the primary parameter to assess the efficiency of the proposed GBCS approach. In this study, it was clearly shown that the use of genetic operators helps avoid premature convergence in the fitness function.

$$\text{Convergence} = \min(i) \text{ such that } \text{Fitness}_i \approx \text{Fitness}_{optimal}$$

The results of the experimental trials carried out on the PMLP dataset show that the baseline Binary Cuckoo Search algorithm converges to a local optimum by iteration 85. However, the GBCS algorithm with crossover and mutation operators achieves a higher peak fitness much earlier in the iterations.

$$Difference = 85\, iterations (Baseline) - 50\, iterations\, (GBCS)$$

$$Speedup = 35\, iterations$$

2.5.3 Feature Reduction Ratio (FRR)

Feature Reduction Efficiency measures an algorithm's effectiveness at eliminating irrelevant features from a high-dimensional dataset. This feature is used in this research to compare the proposed GBCS algorithm with the baseline method by comparing the sizes of the final selected feature subsets produced by these algorithms. It is observed that the higher the reduction ratio, the better the performance of the proposed algorithm.

$$Ratio = (1 - \frac{|S|}{|F|}) \times 100\%$$

For the plant leaf datasets, Inception-V3 generates 2,048 feature candidates, denoted as |F|. Subsequently, the GBCS algorithm finds an optimal subset of selected features, denoted |S|. The reduction ratio is calculated as:

$$Ratio = (1 - \frac{|S|}{2048}) \times 100\%$$

2.5.4 Precision, Recall, and F1-Score

To ensure a comprehensive evaluation of the classifier's performance, especially given the 1:1.71 class imbalance ratio inherent in PMLP, the F1-score is used as the performance metric. The F1-Score is the harmonic mean of Precision and Recall, such that the performance of the classifier is consistent for all 40 classes of plants and does not bias towards some of the dominant species like Tamarindus indica.

Precision Calculation:

$$Precision = \frac{TP}{TP+FP}$$

$$Precision = \frac{23}{23 + 5}$$

$$Precision = \frac{23}{28}$$

$$Precision = 0.8214$$

Recall Calculation:

$$Recall = \frac{TP}{TP+FN}$$

$$Recall = \frac{23}{23 + 2}$$

$$Recall = \frac{23}{25} = 0.9200$$

F-1 Score Calculation:

$$F1 = 2 \times \frac{Precision \cdot Recall}{Precision + Recall}$$

$$F1 = 2 \times \frac{0.8214 \cdot 0.9200}{0.8214 + 0.9200},$$

$$F1 = 2 \times \frac{0.7557}{1.7414}$$

$$F1 = 0.8679$$

2.6 Phase 6: Deployment

In the final phase, the GBCS algorithm selects the optimal feature set, which is used to develop the predictive model. The final phase can be considered the culmination of the study, resulting in a refined model that efficiently classifies new images of the leaf while ensuring minimal computational costs. The classifier developed can classify 40 medicinal plant species found naturally in the Philippines.

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